Parag Sharma

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Professional Summary

Final-year CSE student specializing in AI/ML — with hands-on experience in Computer Vision, RAG pipelines, and LLM application development using LangChain, LCEL, and HuggingFace. International hackathon winner recognized by Taiwan's Ministry of Digital Affairs for building edge-optimized real-time CV systems. Stanford ML Specialization certified, published patent co-author, and open-source contributor. GenAI educator to 150+ students via Google Developer Groups.

Honors & Awards

- **Good Idea Award Winner (Taiwan):** Won $40,000 NT prize in an international hackathon by the Ministry of Digital Affairs, Taiwan for AI-powered accessibility innovation. *Nov 2024*
- **Top 5 National Finalist:** Ranked Top 5 out of 3000+ teams in Hack4Change organized by Google and Charcha. *Aug 2024*
- **National Innovation Award Runner-Up:** Secured 2nd position in NIA'24 organized by P.I.E.T. *Nov 2024*
- **2nd Runner-Up, Ideathon 2025:** Awarded 3rd prize for AI innovation at Ambala College of Engineering. *Feb 2025*

Technical Skills

Languages	Python, C++, SQL, JavaScript
ML / DL	Scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib, XGBoost
Computer Vision	OpenCV, MediaPipe, Hand/Pose Landmarker, Edge Inference
LLM & GenAI	LangChain (LCEL), RAG, FAISS, ChromaDB, HuggingFace Transformers, Groq API, OpenAI API, Vertex AI (Gemini), Prompt Engineering
Cloud & Tools	Google Cloud Platform, Streamlit, Hugging Face Spaces, Salesforce Einstein AI, Git/GitHub, Jupyter, VS Code

Projects

Silent Voice — Real-Time Sign Language Interpreter *Oct 2024 – Dec 2024*

Tech: Python, OpenCV, MediaPipe, Edge Inference, OS Audio APIs GitHub

- **International Winner (Taiwan):** Built a real-time ASL-to-speech system enabling communication between hearing-impaired users and non-signers, winning the Ministry of Digital Affairs hackathon out of global submissions.
- Designed an edge-optimized CV pipeline using **OpenCV** and **MediaPipe Hand Landmarker** to extract 21-point finger geometry per frame, achieving real-time inference at **30+ FPS** on CPU-only hardware.
- Mapped extracted landmarks to an ASL dictionary of 26 characters and triggered audible translations via native OS audio APIs — zero cloud dependency, fully offline.

RepoSage — AI Codebase Intelligence Tool *Mar 2026 – Apr 2026*

Tech: Python, LangChain (LCEL), ChromaDB, HuggingFace Embeddings, Groq API, Streamlit GitHub

- Implemented **multi-hop retrieval** — a two-stage RAG strategy that retrieves top-k semantically relevant code chunks, then performs a second fetch across all related files identified in hop-1 metadata, simulating how a developer navigates connected modules rather than isolated snippets.
- Built a **code-aware ingestion engine** that parses GitHub repositories and persists structured chunks in **ChromaDB** with rich metadata (filepath, language, chunk type, start line); supports multiple repos simultaneously via named collections with content deduplication and 10-chunk context-window cap.
- Engineered a production-style **LangChain LCEL chain** (`RunnablePassthrough | Prompt | Llama 3.1 | StrOutputParser`) with source file attribution; deployed via **Streamlit** with private repo support through GitHub token authentication.

Orbcura — Accessibility AI App *Jul 2024 – Sep 2024*

Tech: Python, OpenCV, MediaPipe Pose Landmarker, Flutter, USSD GitHub

- Built an assistive application for visually impaired users integrating **real-time environment recognition** and accessible digital payments.
- Developed a real-time activity-monitoring module using **MediaPipe Pose Landmarker** — applying kinematic joint analysis and spatial distance calculations to track and interpret user movements in real time.
- Engineered a USSD-based UPI payment layer bypassing internet/permission constraints; the offline accessibility workflow resulted in a **co-authored published patent**.

Open-Source Contributions

- **TheAlgorithms/Python** (217k+ stars): Implemented an O(n) Sliding Window Maximum using Monotonic Deque; PR reviewed, approved, and merged into main.

- **Zomato/espresso**: Authored production-grade documentation for ZeroLogger integration and observability best practices.
- **Agentscope/Trinity-RFT**: Improved contributor onboarding by adding a CPU-only quick-start guide and clarifying hardware requirements for LLM training.

Experience

Software Engineer Intern *May 2024 – Jun 2024*

TriColor Initiatives Pvt. Ltd. | Panchkula, Haryana

- Gained hands-on exposure to **Salesforce Einstein AI** for predictive analytics and AI-driven automation within enterprise Salesforce ecosystems.
- Developed encrypted REST APIs reducing third-party integration time by **20%**; optimized database reporting workflows improving query performance by **25%**.
- Built backend automation deployed to **500+ active users**; implemented Connected App security for controlled third-party integrations.

Leadership & Community

Community Lead & Organizer *Sept 2024 – Jul 2025*

Google Developer Groups (GDG) on Campus | Mullana, Haryana

- **Hackathon Lead:** Organized national-level events (GDG Hackureka, GDG Hacks), coordinating **400+ participants** and **150+ teams** across India.
- **GenAI Educator:** Designed and delivered hands-on workshops on Generative AI using **Google Cloud Vertex AI and Gemini APIs** for 150+ students.
- Led cross-functional student engineering teams using Agile workflows to deliver large-scale technical events end-to-end.

Education & Certifications

B.Tech, Computer Science & Engineering | CGPA: 8.03 *May 2026*

Maharishi Markandeshwar (Deemed To Be University) | Mullana, Haryana

Certifications:

- *Machine Learning Specialization* — Andrew Ng, Stanford / DeepLearning.AI (Coursera) *Sept 2023*
- *Salesforce AI Associate* *June 2024*